15 One end of a spring is fixed to a support. A block of mass 2.0 kg is attached to the other end of the spring, which causes the spring to extend by 5.0 cm.

The block is then pulled down a distance of 2.5 cm and released to perform a simple harmonic motion of period 0.45 s.

What is the amplitude and angular frequency of the motion?

	amplitude / cm	angular frequency / rad s^{-1}
A	7.5	0.45
B	7.5	14
C	2.5	0.45
D	2.5	14

Ans: D
The displaced amount = amplitude of oscillation = 2.5 cm

$\omega = 2\pi/T$

16 Which of the following shows oscillation of decreasing order of damping?